



IC 253: Programming and Data Structures

Course Code	IC 253
Course Name	Programming and Data Structures
L-T-P-C	2-0-2-3
Prerequisites	None
Intended for	B.tech (1 st Yr)
Mutual Exclusion	None
Approval	54th BoA (May 2024)

Course Contents

- **Introduction:** Review of problem-solving using computers; Importance of data structure and algorithms; Elementary data structures: array, linked lists, stack and queues; Abstract data Operations on elementary data structures; Time and space complexity of algorithms: asymptotic analysis and notation, average, and worst-case analysis. Subtopics (3 Hours)
- **Stack and Queues:** Sequential and linked implementations, representative applications such as towers of Hanoi, and parenthesis matching. (3 Hours)
- **Lists:** Abstract data type, sequential and linked representations, comparison of insertion, deletion and search operations for sequential and linked lists, list and chain classes, doubly linked lists, circular lists, skip lists, applications of lists in sparse tables. (3 Hours)
- **Trees:** Abstract data type, sequential and linked implementations, tree traversal methods and algorithms, Binary trees and their properties. (3 Hours)
- **Search Trees:** Binary search trees, search efficiency, insertion, and deletion operations, the importance of balancing, AVL trees, searching, insertion, and deletions in AVL trees. (4 Hours)
- **Heaps:** Heaps as priority queues, heap implementation, insertion-deletion operations, and heapsort. (2 Hours)
- **Graphs:** Definition, terminology, directed and undirected graphs, properties, implementation - adjacency matrix and linked adjacency chains, connectivity in graphs, graph traversal - breadth first and depth first, spanning trees. (3 Hours)
- **Basic Algorithm Techniques:** Greedy algorithms, divide & conquer. Search techniques - backtracking, Sorting algorithms with analysis, integer sorting, selection sort. Graph algorithms: DFS and BFS with applications, MST, and shortest paths. (7 Hours)

Laboratory Modules

- Included with each Unit

Text Books

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, **Introduction to Algorithms**, 3rd Edition, MIT Press, 2009.

References

1. S. Sahni, **Data Structures, Algorithms, and Applications in C++**, 2nd Edition, Silicon Press, 2005.
2. A. M. Tenenbaum, Y. Langsam, and M. J. Augenstein, **Data Structures Using C and C++**, 2nd Edition, Prentice Hall, 1995.